ECE 1508: Reinforcement Learning

Course Logistics

Ali Bereyhi

ali.bereyhi@utoronto.ca

Department of Electrical and Computer Engineering

University of Toronto

Fall 2026

Welcome to ECE 1508!

Great pleasure to see you in ECE 1508

Reinforcement Learning

Instructor: Ali Bereyhi

- *Office:* BA 7208 at Bahen Centre for Information Technology
- *Email:* `ali.bereyhi@utoronto.ca`

Where and When?

- *Thursdays at 1:00 – 4:00 PM* at **BA-1170**

Tutorials

- *Fridays at 5:00 PM till 6:00 PM* at **BA-1130**

Instructional Team

- **Dr. Erfan Meskar** – *Project Coordination*
  - Postdoctoral Fellow, ECE Department
  - *Email:* `e.meskar@utoronto.ca`
- **Eason Qu** – *Tutorials*
  - PhD Candidate, ECE Department
  - *Email:* `eason.qu@mail.utoronto.ca`
- **Likun Cai** – *Course Coordination and Office Hours*
  - PhD Candidate, ECE Department
  - *Email:* `likun.cai@mail.utoronto.ca`
- **Mohammadreza Safavi** – *Exam Coordination*
  - MASc Student, ECE Department
  - *Email:* `mohammadreza.safavi@mail.utoronto.ca`

Course Page

Pretty simple and sustainable!

- **GitHub Page.** You get all resources through the GitHub page
- **Quercus.** Used for announcement
- **Crowdmark.** Used for submissions
- **Piazza.** Used for asking questions and searching for team-mates

Please use Piazza!

Please initiate a Piazza post whenever you got a question

↳ Use the right topic label to be safe

If you want to send an email . . .

1. Are you sure? ☺
2. You're sure! Please add the prefix `[ECE1508:RL]` to subject

What Do We Learn?

In nutshell: we learn Reinforcement Learning!

You may wonder how do we learn it? Well! in 3 steps

- *Step 1:* **Fundamentals of Reinforcement Learning**
  - We try to get understand the underlying framework
  - We understand what the main problem is
    - ↳ *We see that it's a sort of optimization problem*
  - We get to look at some simple example

What Do We Learn?

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- *Step 2:* **Reinforcement Learning Methods**
  - Model-based Methods
    - *In some toy-scenarios, we can write the underlying problem analytically*
    - *This is not really the case in practice though!*
  - Model-free Methods
    - *In reality, we cannot write the problem analytically!*
    - *We need to solve the problem directly from data by efficient algorithms*

At this point

- ✔ You can formulate a Reinforcement Learning problem
- ✔ You can implement an algorithms to solve it
- ✘ *But! Your algorithm might take for ever to run!* ☺

What Do We Learn?

In nutshell: *we learn Reinforcement Learning!*

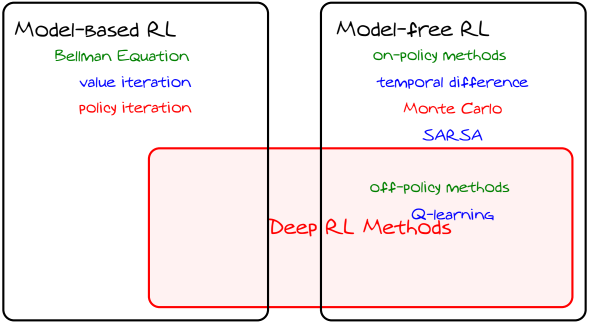
You may wonder how do we learn it? Well! in 3 steps

- *Step 3:* **Deep Reinforcement Learning**
  - We now apply deep learning to solve those hard problems
    - *We use neural networks to learn the solution from few samples*
    - *We look into Deep Q-Learning and Policy Gradient Methods*
    - *We also learn actor-critic methods*

This is the major part of the course $\approx$ *50%*

- You need background on Machine Learning

High Level Chart



We will see this chart couple of times in this course!

How Do We Get Trained?

1. **Assignments.** It comprises 40% of coursework
   - Four sets of assignments
   - Both written and programming questions
     - ↳ Each assignment has two weeks time
     - ↳ *No late submission is accepted!* Sorry!
   - You also get *Practice Questions*
     - ↳ Not submitted anywhere!
     - ↳ For further practice and exam preparation only

Use of AI for Assignments

AI may be used in an extremely restricted fashion

↳ *clarifying concepts, debugging errors, performing sanity checks*

Copying AI responses or relying on AI design/code/etc is *prohibited*

↳ *Complete **AI Usage Attestation** for every submission*

We try to inspect submissions for potential AI usage

How Do We Get Trained?

2. **Exams.** Two short exams comprising 30% of coursework
   - First exam on **October 23**
   - Second exam on **December 04**
   - Hold on tutorial hour, i.e., Friday 5:00-6:00 PM
   - Each exam contains two questions
     - ↳ *No detailed programming is asked*
     - ↳ *Similar to practice questions and assignments*

How Do We Get Trained?

3. **Final Project.** It comprises 30% of coursework
   - Groups of 3. *Smaller or large groups are generally not accepted*
   - Submit proposal by **October 12**
     - ↳ *Both research and design projects are accepted*
     - ↳ *Git repository must be initiated and shared with proposal*
   - Submission on the **second and third weeks of December**
     - ↳ *5-minute presentation the second week, i.e., December 7 – 11*
     - ↳ *Final report and code submission by the end of third week, i.e., December 18*

Use of AI for Assignments

AI may be used in an intelligent assistant for tasks like

↳ *clarification, background, code modules, benchmark implementations, debug, polishing documentation, etc*

Delegating substantial portions to AI, or AI-written report is *prohibited*

↳ *Complete* ***AI Usage Attestation*** *for final submission*

Lots of Programming in *Python*

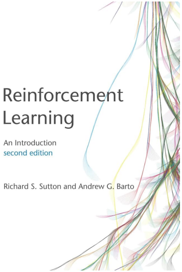
We do programming in Python

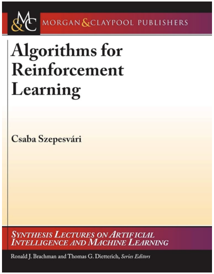
- ↳ We use *PyTorch* and *NumPy* a lot
- ↳ It's important to mention *knowing Machine Learning is a must*!
  - ↳ You may follow without Deep Learning till midterm
  - ↳ After midterm we need to use deep learning
- ↳ For environment implementation, we use
  - ↳ *Gymnasium:* standard API widely used for Reinforcement Learning

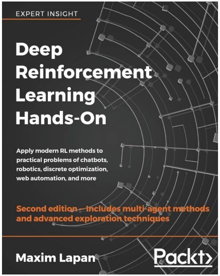
Course Calendar

Week #	Date	Notes	Posted	Deadlines
1	*Sep 07 - Sep 11*			
2	Sep 14 - Sep 18		Assignment 1	
3	Sep 21 - Sep 25			
4	Sep 28 - Oct 02			Assignment 1
5	Oct 05 - Oct 09		Assignment 2	
6	Oct 12 - Oct 16			**Project** *Proposal*
7	Oct 19 - Oct 23	***Exam I on Oct 23***		Assignment 2
8	Oct 26 - Oct 30	***Reading Week-- No Lectures***		
9	Nov 02 - Nov 06		Assignment 3	
10	Nov 09 - Nov 13			
11	Nov 16 - Nov 20		Assignment 4	Assignment 3
12	Nov 23 - Nov 27			
13	Nov 30 - Dec 04	***Exam II on Dec 04***		Assignment 4
14	Dec 07 - Dec 11			
15	Dec 14 - Dec 18	***Examination Time -- No Lectures***		***Project Presentation, Report, and Codes***

Textbooks



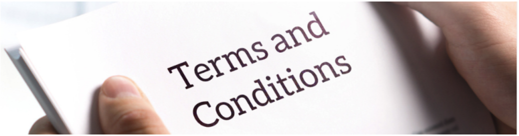




All materials are provided in the course. It's however good to know some texts!

- Sutton abd Brato can be accessed online at this link
- Szepesvári is available online here
- Lapan is a good source for Part 3

Terms and Conditions!



The instructor keeps the right reserved for himself to modify the slides

- *last minute before the lecture* ☺
- *after the lecture has been given*
  - *Typically happens due to typos*

The instructor keeps the right reserved for himself to deliver the lecture-notes

- *in form of mini-batches* ☺

Date and Signature ....

No such thing as a stupid question!

Did you know that we got a Wikipedia page on this?

↳ Trust me! Your question will never sound stupid!

↳ If you don't ask; then, I need to ask!

 ↳ Interaction is the best tool to avoid getting bored!

Any Questions? ☺